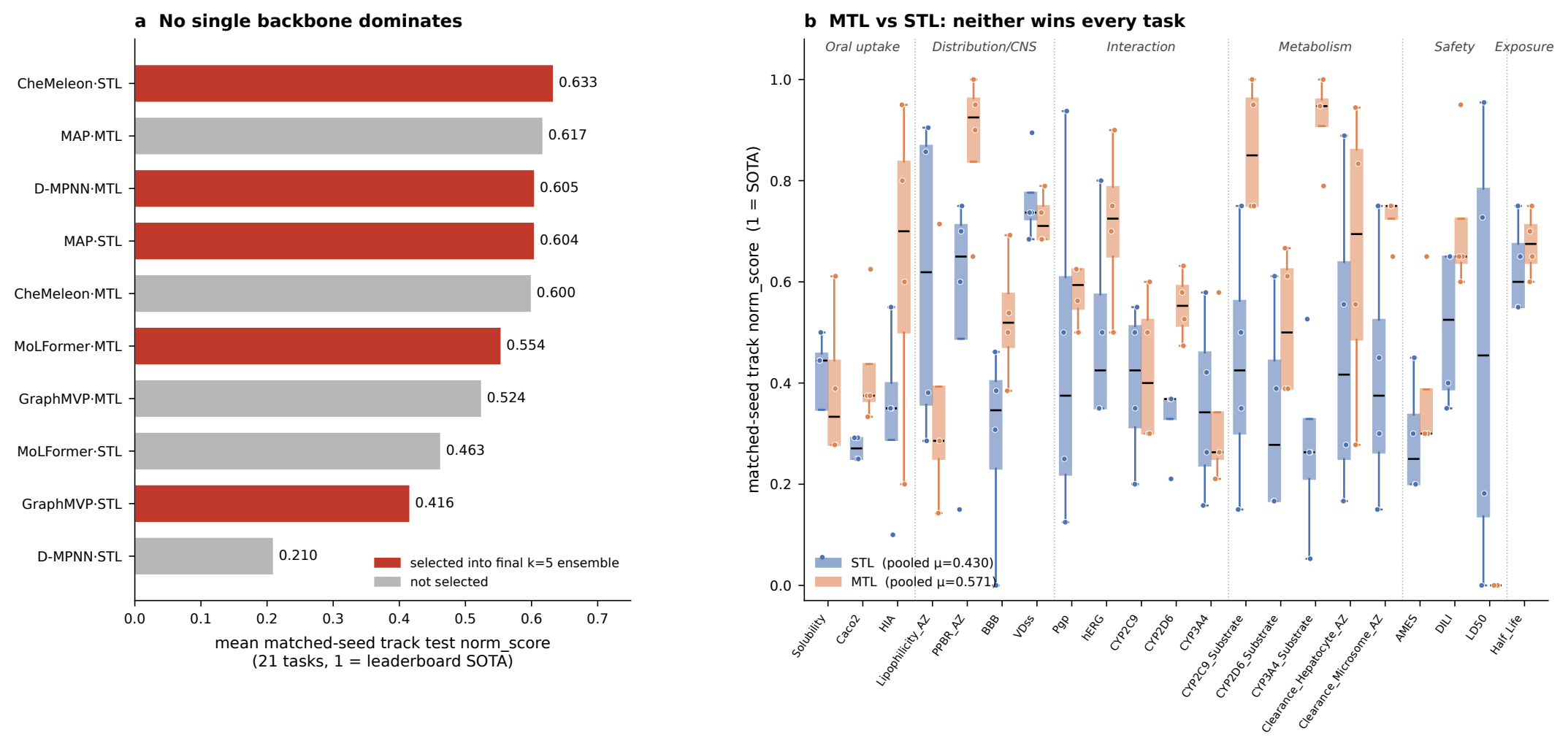

a No single backbone dominates
CheMeleon·STL
MAP·MTL
D-MPNN·MTL
MAP·STL
CheMeleon·MTL
MoLFormer·MTL
GraphMVP·MTL
MoLFormer·STL
GraphMVP·STL
D-MPNN·STL
0.633
0.617
0.605
0.604
0.600
0.554
0.524
0.463
0.416
0.210
selected into final k=5 ensemble
not selected
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
mean matched-seed track test norm_score
(21 tasks, 1 = leaderboard SOTA)
b MTL vs STL: neither wins every task
Oral uptake
Distribution/CNS
Interaction
Metabolism
Safety
Exposure
matched-seed track norm_score (1 = SOTA)
1.0
0.8
0.6
0.4
0.2
0.0
STL (pooled μ=0.430)
MTL (pooled μ=0.571)
Solubility
Caco2
HIA
Lipophilicity_AZ
PPBR_AZ
BBB
VDss
Pgp
hERG
CYP2C9
CYP2D6
CYP3A4
CYP2C9_Substrate
CYP2D6_Substrate
CYP3A4_Substrate
Clearance_Hepatocyte_AZ
Clearance_Microsome_AZ
AMES
DILI
LD50
Half_Life